HUNADI SYLVIAH MOTUBATSE

SQL Practical Excercise 4:
Joins, Union, Filtering and Aggregation

JOINS

Question 1

```
SELECT *
FROM Employees As A
INNER Join Projects As B
ON A.Employee_ID = B.Employee_ID;
```

Emp-ID	FirstName	LastName	Depart	Salary	Project-ID	ProjNam	Budget	Stat
1	John	Doe	IT	70000	101	AIDev	10000	comp
1	John	Doe	IT	70000	108	cyb Audit	75000	Pend
2	Alice	Smith	HR	60000	102	Emp Traing	50000	on going
3	Bob	Johnson	Finance	75000	104	Financial Training	90000	on
5	Emma	Wilson	Sales	65000	105	market Expan	65000	Comp
6	Micheal	Clark	Finance	80000	106	Risk Manager	80000	Pend

Question 2

```
SELECT *
FROM Employees As A
LEFT JOIN Projects As B
ON A.employeesID=A.Employees_ID;
```

Emp-ID	FirName	LastName	Depart	salary	ProjectID	ProjName	Budget	status
1	John	Doe	IT	70000	101	AIDev	10000	complete
1	John	Doe	IT	70000	103	Cyb Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Emp Traing	50000	Ongoing
3	Bob	Johnson	Fina	75000	104	Fin Traing	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	sales	65000	105	market Exp	65000	complete
6	Michael	Clark	Finance	80000	106	Risk mnt	80000	Pending

JOINS

Question 3

```
SELECT*
FROM Project AS A
Right Join Employee AS B
```

Proj-ID	Pro_NAme	Budget	Status	EmpID	FirName	LasName	Department	Salary
101	AI Dev	100000	Complete	1	John	Doe	IT	70000
103	Cyb Audit	75000	Pending	1	John	Doe	IT	NULL
102	EmploTrai	50000	Ongoing	2	Alice	Smith	HR	60000
104	Financial	90000	ongoing	3	Bob	Johson	Finance	95000
105	Market Exp	65000	complete	5	Emma	Wilson	Sales	65000
106	Risk Mnt	80000	pending	6	Michael	Clark	Finance	80000

Question 4

```
SELECT*
FROM Employee AS A
FULL JOIN Project AS B
```

EmID	FirName	LasNam	Depart	Salary	ProName	ProjeID	Budget	Status
1	John	Doe	IT	70000	AI Dev	101	10000	Comple
1	John	Doe	IT	NULL	Cyb Audit	103	75000	Pending
2	Alice	Smith	HR	60000	EmpTraing	102	50000	Ongoin
3	Bob	Johson	Finance	75000	Financial Analysis	104	90000	ongoin
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	sales	65000	Expan Market	105	65000	Complete
6	Michael	Clark	Finance	80000	Risk Mnt	106	80000	Pendin

2 UNION & UNION ALL

5. To UNION Two tables the column must be the same number of column on both tables
If Not similar New column must be generated

```
SELECT City as Location
FROM Employees
Union
SELECT Status as location
From project;
```

Results with No duplication:
Location
New york
Los Angel
Toronto
London
Sudney
Completed
Ongoing
Pendinding

6. SELECT City AS Location
FROM Employees
UNION ALL
SELECT Status AS Location
FROM project;

Location (Returns All Including duplic
New york
Los Angeles
Toronto
London
Sydney
New york
Completed
On going
Pending
On going
Completed
Pending

3 Filtering Statement

7. SELECT Employee-ID,
Employee-FirstName,
Employee-lastName,
Department,
Salary
FROM employee
WHERE Salary > 70000;

EMID	FirstN	LastN	Depart	Salary
3	Bob	Johnson	FINANCE	75000
4	David	Brown	IT	72000
5	Micheal	Clark	FINANCE	80000

8. SELECT Emp-ID,
FirstName,
LastName,
Departmen,
Salary
FROM Employee
WHERE Department = 'IT' OR Department = 'Finance';

EMD-ID	FirstName	lastName	Department
1	John	Doe	IT
4	David	Brown	IT
3	Bob	Johnson	Finance
6	Michael	clark	Finance

9. SELECT project_ID,
Project_Name,
Project - Budget,
Status
FROM project
WHERE status != 'Completed';

Proj-ID	ProjeName	Budget	status
103	Cyber Audit	75000	Pending
10[illegible]	Emp Training	50000	On going
104	Fin Analysis	90000	On-going
106	Risk Mnt	80000	Pending

10. SELECT PROJID,
ProjectName,
Budget,
Status
FROM project
WHERE Status != 'Completed' and Salary > 70000;

Project-ID	ProjectName	Budget	Status
103	CyberAudit	75000	Pending
104	Fin Analysis	90000	ongoing
106	Risk Mnt	80000	Pending

11 SELECT Emp-ID,
FirstName
Lastname, Departments
Salary
FROM Employees
WHERE city = 'New york' OR City = 'Toronto'
ORDER By SALARY DECS;

Emp-ID	FirstName	lastName	Depart	Salary
1	John	Doe	New york	80000
3	Bob	Johnson	Toronto	75000
6	Michael	Clark	New york	70000

12. SELECT EMp-ID
FirstName
LastName
Department
Salary;
FROM employees
ORDER By salary desc
Limit 3;

Emp-ID	FirstName	LastName	Department	Salary
6	Michael	Clark	Finance	80000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000

4. Aggregation Function with Group By & Having statement

13 SELECT Department
Sum (salary) As Total-salary
FROM employees
Group By Department
ORDER By Total-salary desc;

Department	Total Salary
Finance	155000
IT	142 000
Marketing	65000
HR	60000

14. SELECT City
AVG (salary) AS AverageSalary
FROM Employees
GRoup By City
Having AverageSalary > 65000;

City	AverageSalary
Newyork	75 000
Toronto	75 000
London	72 000

15 SELECT Department,
COUNT(EmpID) AS EmployeeCount
FROM Employees
GROUP By Department
Having EmployeeCount >1;

Department	Employeecount
IT	2
Finance	2

16. SELECT Status,
COUNT(ProjectID) AS ProjectCount
FROM project
GROUP by Status
Having projectCount >=2;

Status	ProjectCount
Pending	1
Completed	1
	2

17. SELECT Emp-ID,
FirstName,
Lastname
Sum (budget) AS TotalprojectBudget
FROM project
GROUP By Emp-ID
Having TotalprojectBudget > 150000;

Emp-ID	FirstName	LastName	TotalprojectBudget
1	John	Doe	175000